

Lead Management System (LMS) - Frontend Engineering Document

Purpose:

This document defines the complete frontend architecture, technology stack, folder structure, components, pages and development rules for building a production-grade React CRM application.

Frontend Framework:

React.js with Vite

1. Frontend Technology Stack

Core:

- React.js (Vite)
- JavaScript / TypeScript Ready Architecture

Routing:

- React Router

State Management:

- Redux Toolkit
- Context API

Server State:

- React Query

API Communication:

- Axios

UI Framework:

- Material UI

Forms:

- React Hook Form

Validation:

- Yup

Tables:

- React Table

Charts:

- Chart.js

Notifications:

- React Toastify

2. Frontend Architecture

Recommended Structure:

src/

components/
Reusable UI components

pages/
Application screens

layouts/
Dashboard layouts

services/
Business services

api/
Axios configuration

redux/
Global state management

hooks/
Custom reusable hooks

context/
Application context

routes/
Protected and public routes

assets/
Images, icons and static files

3. Authentication Flow

Features:

- Login Page
- Token Storage
- Protected Routes
- Role Based UI Rendering
- Auto Logout
- Permission Checking

Flow:

User Login
↓
API Authentication
↓
Store Token + User Role
↓

Redirect Dashboard



Control Access using RBAC

4. Application Pages

Public Pages:

Login

Private Pages:

Dashboard

Lead List

Add Lead

Edit Lead

Lead Details

Follow-ups

Reports

Users Management

Profile

Settings

5. Dashboard Requirements

Dashboard Cards:

- Total Leads
- Today's Leads
- New Leads
- Contacted Leads
- Follow-ups
- Qualified Leads
- Won Leads
- Lost Leads
- Conversion Rate

Charts:

Lead Source Chart

Monthly Leads Chart

Sales Performance Chart

Conversion Analytics

6. Lead Management UI

Lead List Features:

- Search
- Advanced Filters
- Pagination
- Sorting
- Bulk Selection
- Export Actions

Lead Form Fields:

Name
Company
Email
Phone
Address
Industry
Source
Campaign
Product
Budget
Priority
Status
Assigned User
Follow-up Date
Notes
Tags
Documents

7. Lead Details Page

Sections:

Overview

Timeline

Follow-ups

Notes

Documents

Emails

Activity Logs

Purpose:

Provide complete 360-degree lead information.

8. Component Design System

Reusable Components:

Button

Input

Select

Modal

Sidebar

Navbar

Cards

Data Table

Charts

Badges

Status Labels

Forms

Loader

Empty State

Error State

9. API Integration Pattern

All APIs should be managed from services.

Example:

api/
axios.js

services/
leadService.js
authService.js
userService.js
reportService.js

Components should never directly call APIs.

10. State Management Rules

Global State:

Authentication

Current User

Permissions

Theme

Server State:

Leads

Reports

Dashboard Analytics

Follow-ups

11. Role Based Frontend Rules

Super Admin:
Full Access

Admin:
Management Access

Sales Manager:
Team Management

Sales Executive:
Assigned Leads Only

Marketing:
Lead Creation

Viewer:
Read Only

12. Production Requirements

Frontend must include:

- Clean folder architecture
- Reusable components
- Responsive design
- Error handling
- Loading states
- Empty states
- Secure routing
- Optimized performance
- Maintainable code